Mastering Generative AI & LLM Engineering
From Foundations to Applications
Lead by : Shreyas Shukla
in association with
iHUB DivyaSampark, IIT Roorkee & RBPL

LangChain

Mastering Generative AI & LLM Engineering
From Foundations to Applications

Large Language Models

Led by : Shreyas Shukla

in association with
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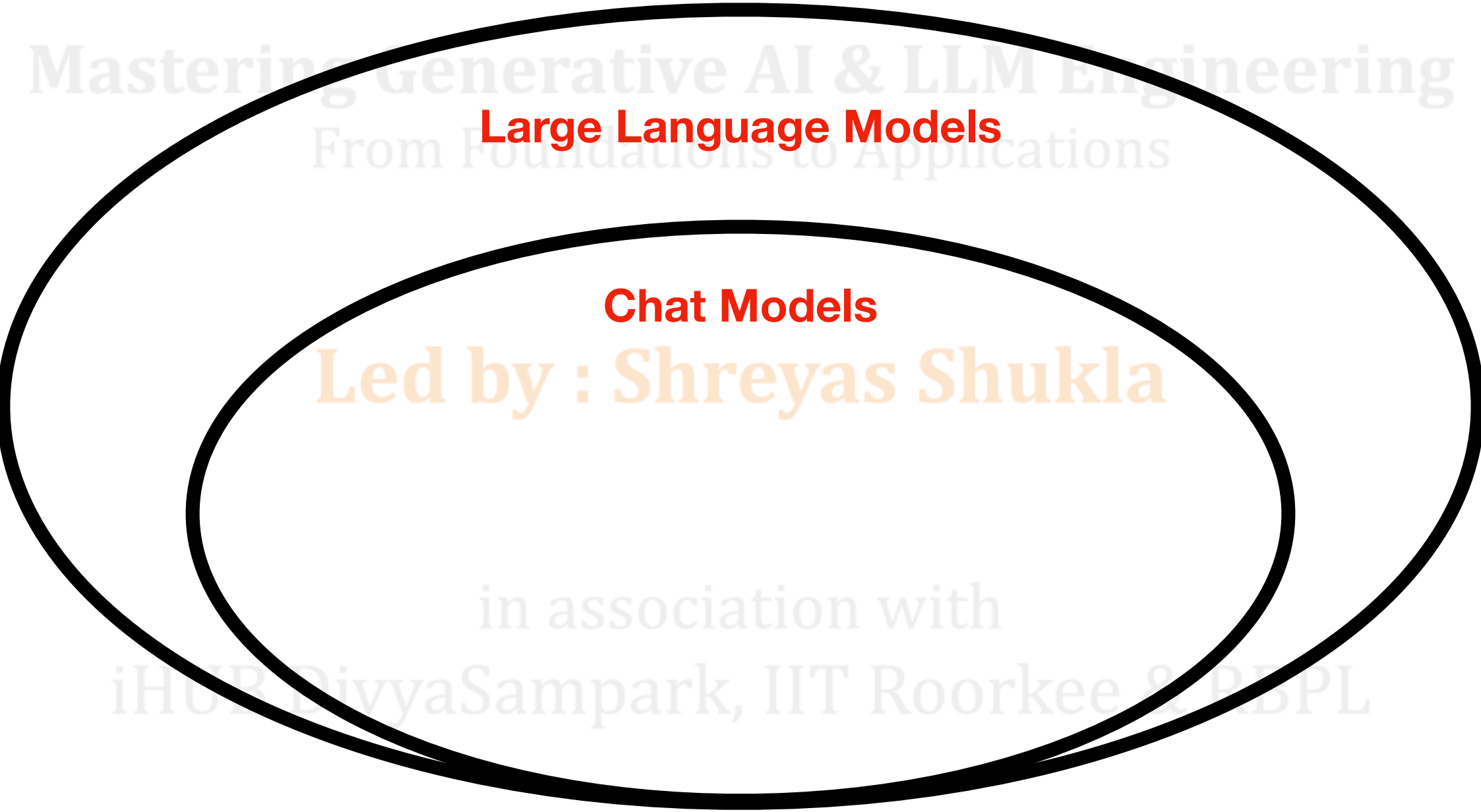

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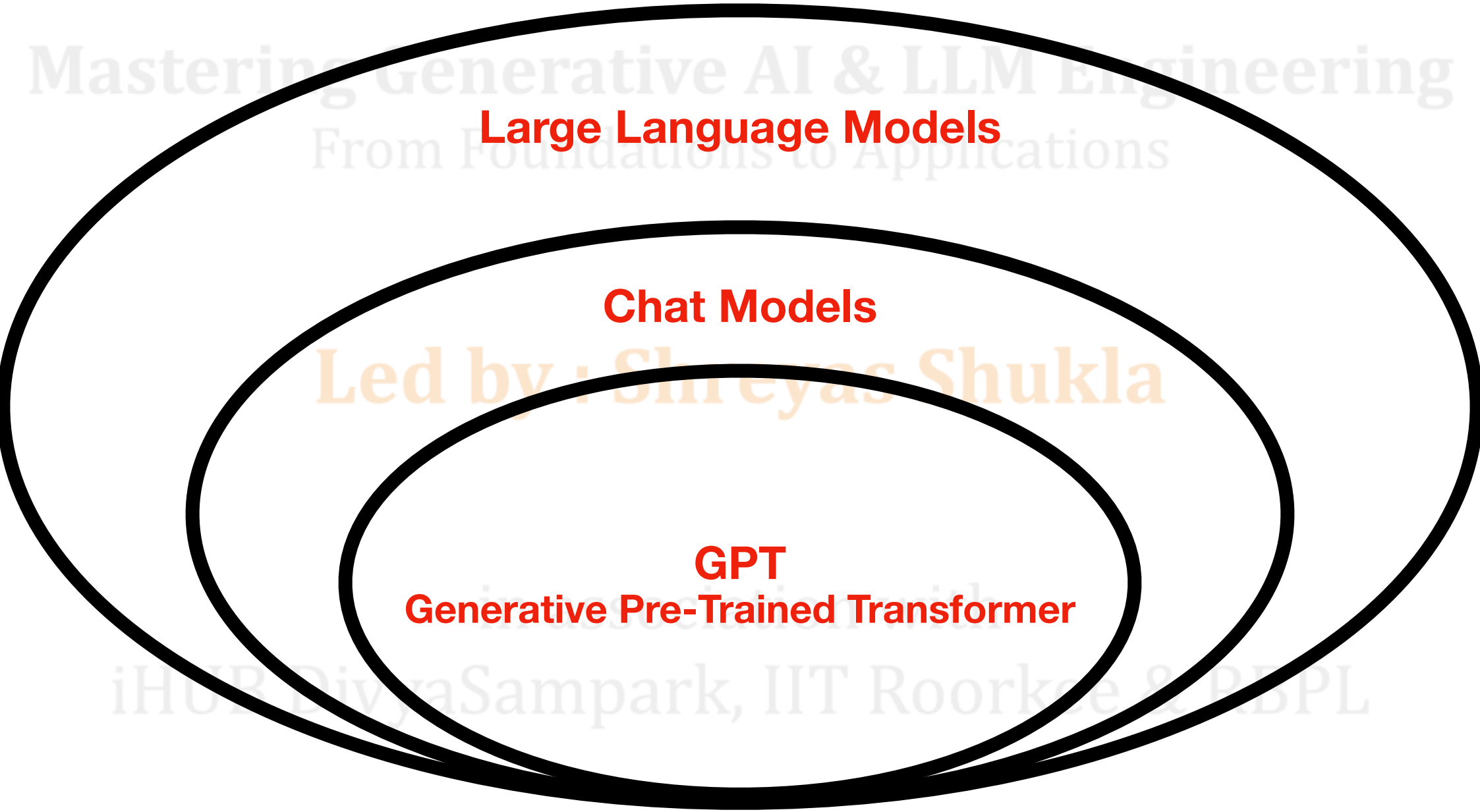

Mastering Generative AI & LLM Engineering
From Foundations to Applications

LangChain

Framework that helps facilitate the integration of LLMs into applications

- Stateful : Keeping Memories of past conversations
- Context-aware : Answering questions about information outside the model's training data
- Reasoning : Choose among various tools

LangChain

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Mastering Generative AI & LLM Engineering

HDFC BANK

Led by : Shubham Shukla

+

LangChain

HDFC Assist

Call Transcript

abcde1234f

Middle Layer

****

LLM

LLM

Automatically summarizes support calls

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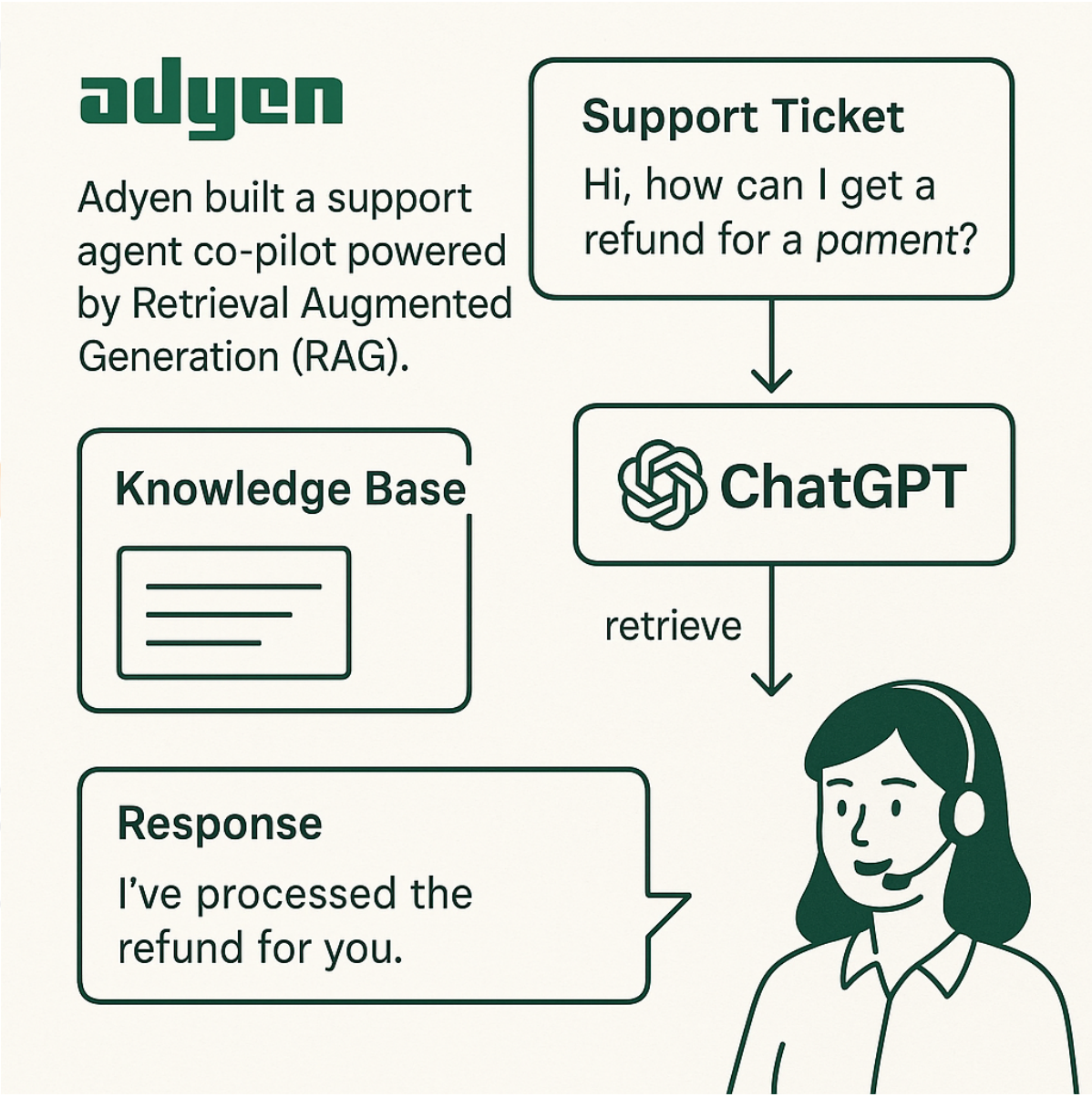

adyen
Adyen built a support agent co-pilot powered by Retrieval Augmented Generation (RAG).
Support Ticket
Hi, how can I get a refund for a *pament*?
ChatGPT
retrieve
Knowledge Base
Response
I've processed the refund for you.

RoboCop & Remark: AI-Powered Python Automation

BEFORE:
Manual Scripting Challenges

Data Extraction | Accounting | HR | Finance | IT

Complex, time-consuming, error-prone.

THE SOLUTION:
RoboCop & Remark Co-pilot

"Scrape product prices from this site"

"Generate monthly invoice summary"

RoboCop Platform (Python)

Remark Co-pilot (LangChain-based)

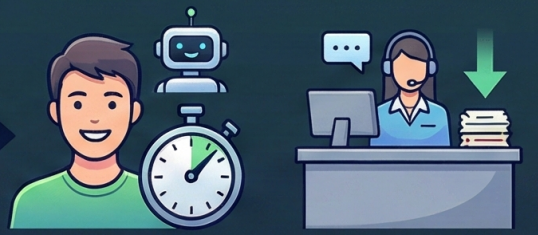
LangChain

Knowledge Base (Examples, Docs, Workflows)

Generated Bot

Describe goal, get code. Trained on real-world examples.

AFTER:
Accelerated Results



Faster Solutions

Fewer Support Tickets

Increased efficiency, reduced support burden.

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Building Chat Applications

With OpenAI and LangChain

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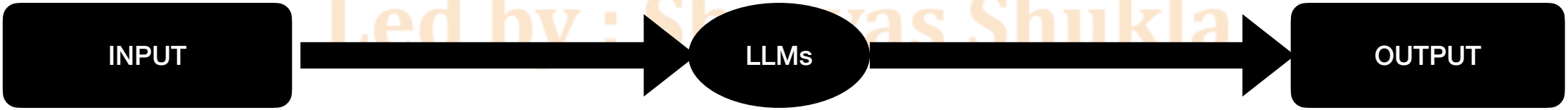

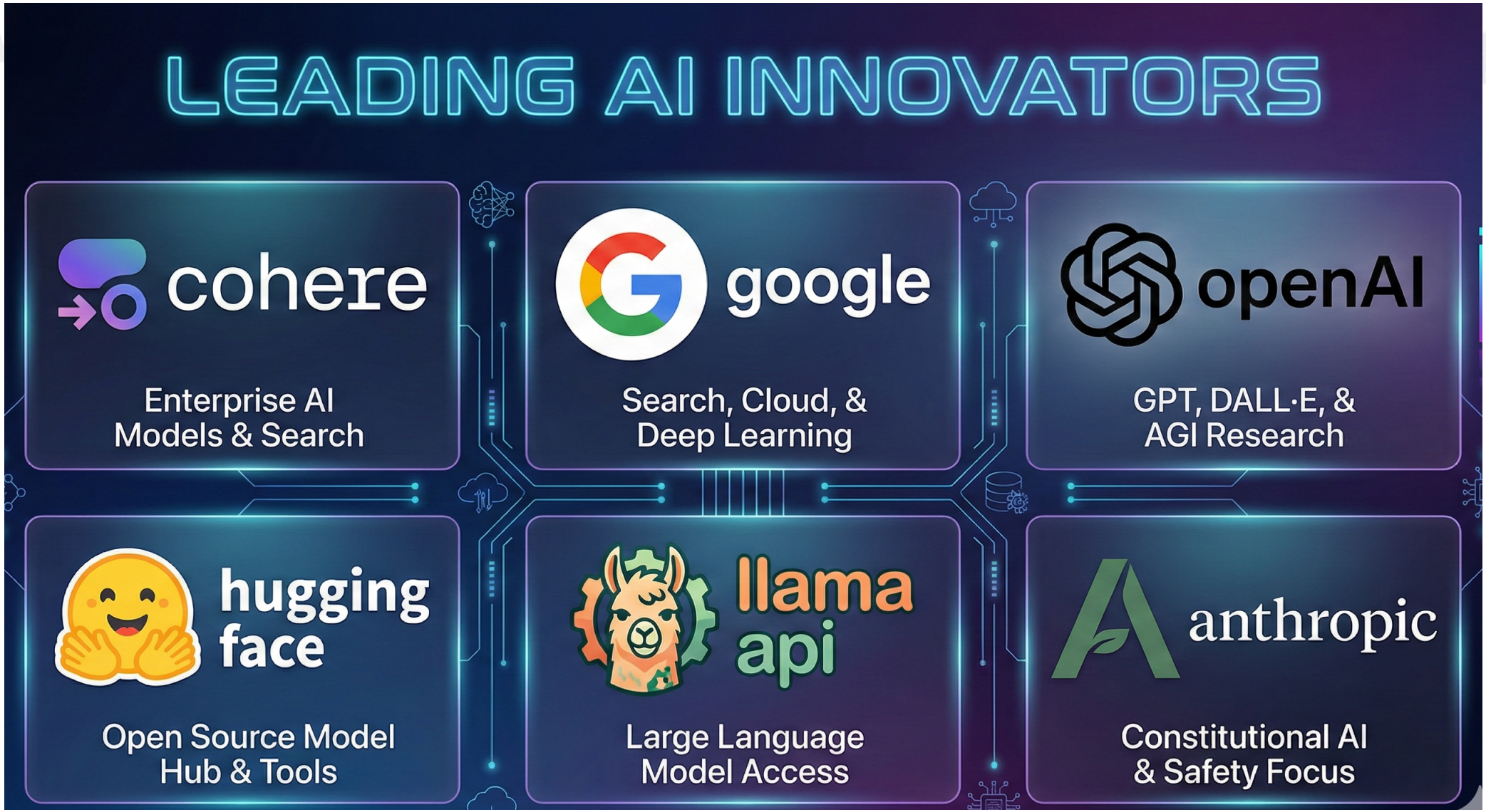
LEADING AI INNOVATORS
cohere
Enterprise AI
Models & Search
google
Search, Cloud, &
Deep Learning
openAI
GPT, DALL·E, &
AGI Research
hugging
face
Open Source Model
Hub & Tools
llama
api
Large Language
Model Access
anthropic
Constitutional AI
& Safety Focus

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Development
Observability
Deployment
LangChain
LangSmith
LangServe

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LangChain continued..

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From Foundations to Applications

LangChain

1. Stateful
2. Context-aware
3. Reasoning

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From Foundations to Applications

Model I/O Module

1. Model
2. Prompt and Prompt Templates
3. Output Parser

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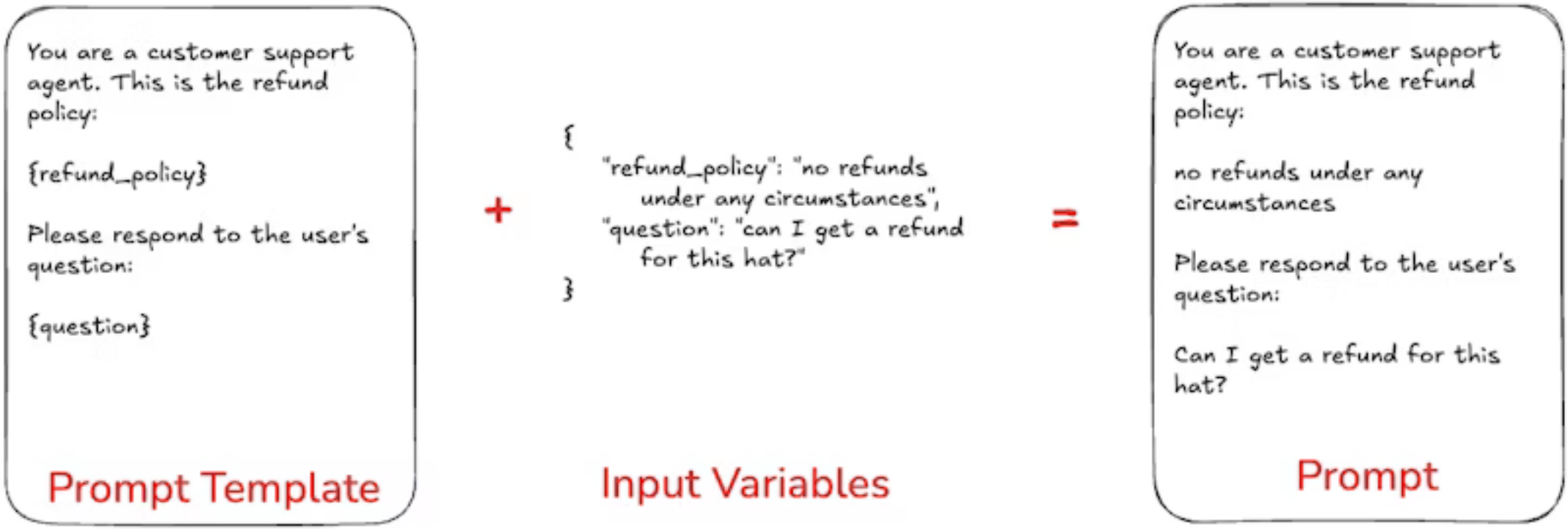

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What about Model Output ?

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What about Model Output ?

OUTPUT PARSERS !!

Because assistant role message unsuitable for python script

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Model I/O Module

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Retrieval Module

1. Document Loader
2. Text Splitter
3. Embedding Model
4. Vector Store
5. Retriever

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Agent Tooling Module

1. Tool
2. Toolkit

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LangChain Expression Language

(Connects all the three modules together)

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- Templates
- LangSmith platform (for product observability)
- LangServe libraries (for deployment)

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Let’s Code !!